METRIC

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SUPERSEDING
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PERFORMANCE SPECIFICATION

CALIBRATING FLUIDS, AIRCRAFT FUEL SYSTEM COMPONENTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for three types of calibrating fluid used in the calibration of aircraft fuel system components.

1.2 Classification. The fluids will be of the following types as specified (6.2):

Type I - Normal Heptane

Type II - Special Run Stoddard Solvent

Type III - High Flash Point Fluid

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AMSC N/A FSC 6850

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-25017 - Inhibitor, Corrosion/Lubricity Improver, Fuel Soluble (NATO S-1747)

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D56 - Standard Test Method for Flash Point by Tag Closed Cup Tester

ASTM D86 - Standard Test Method for Distillation of Petroleum Products and Liquid Fuels at Atmospheric Pressure

ASTM D93 - Standard Test Methods for Flash Point by Pensky-Martens Closed Cup Tester

ASTM D130 - Standard Test Method for Corrosiveness to Copper from Petroleum Products by Copper Strip Test

ASTM D156 - Standard Test Method for Saybolt Color of Petroleum Products (Saybolt Chromometer Method)

ASTM D323 - Standard Test Method for Vapor Pressure of Petroleum Products (Reid Method)

ASTM D381 - Standard Test Method for Gum Content in Fuels by Jet Evaporation

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ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D873	-	Standard Test Method for Oxidation Stability of Aviation Fuels (Potential Residue Method)
ASTM D1093	-	Standard Test Method for Acidity of Hydrocarbon Liquids and Their Distillation Residues
ASTM D1298	-	Standard Test Method for Density, Relative Density, or API Gravity of Crude Petroleum and Liquid Petroleum Products by Hydrometer Method
ASTM D1319	-	Standard Test Method for Hydrocarbon Types in Liquid Petroleum Products by Fluorescent Indicator Adsorption
ASTM D2276	-	Standard Test Method for Particulate Contaminant in Aviation Fuel by Line Sampling
ASTM D2386	-	Standard Test Method for Freezing Point of Aviation Fuels
ASTM D3227	-	Standard Test Method for (Thiol Mercaptan) Sulfur in Gasoline, Kerosine, Aviation Turbine, and Distillate Fuels (Potentiometric Method)
ASTM D3242	-	Standard Test Method for Acidity in Aviation Turbine Fuel
ASTM D3606	-	Standard Test Method for Determination of Benzene and Toluene in Spark Ignition Fuels by Gas Chromatography
ASTM D3828	-	Standard Test Methods for Flash Point by Small Scale Closed Cup Tester
ASTM D4052	-	Standard Test Method for Density, Relative Density, and API Gravity of Liquids by Digital Density Meter
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4177	-	Standard Practice for Automated Sampling of Petroleum and Petroleum Products
ASTM D4952	-	Standard Test Method for Qualitative Analysis for Active Sulfur Species in Fuels and Solvents (Doctor Test)
ASTM D5191	-	Standard Test Method for Vapor Pressure of Petroleum Products and Liquid Fuels (Mini Method)

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ASTM D5304	-	Standard Test Method for Assessing Middle Distillate Fuel Storage Stability by Oxygen Overpressure
ASTM D5452	-	Standard Test Method for Particulate Contamination in Aviation Fuels by Laboratory Filtration
ASTM D5972	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Phase Transition Method)
ASTM D6045	-	Standard Test Method for Color of Petroleum Products by the Automatic Tristimulus Method
ASTM D6379	-	Standard Test Method for Determination of Aromatic Hydrocarbon Types in Aviation Fuels and Petroleum Distillates – High Performance Liquid Chromatography Method with Refractive Index Detection
ASTM D7153	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Laser Method)
ASTM D7154	-	Standard Test Method for Freezing Point of Aviation Fuels (Automatic Fiber Optical Method)
ASTM D7619	-	Standard Test Method for Sizing and Counting Particles in Light and Middle Distillate Fuels, by Automatic Particle Counter
ASTM D7777	-	Standard Test Method for Density, Relative Density, or API Gravity of Liquid Petroleum by Portable Digital Density Meter
ASTM E29	-	Standard Practice for Using Significant Digits in Test Data to Determine Conformance with Specifications

(Copies of these documents are available from https://www.astm.org/.)

ENERGY INSTITUTE

IP 540	-	Determination of the existent gum content of aviation fuel – Jet evaporation method
IP 565	-	Determination of the level of cleanliness of aviation turbine fuel – Portable automatic particle counter method
IP 577	-	Determination of the level of cleanliness of aviation turbine fuel – Automatic particle counter method using light extinction

(Copies of these documents are available from https://publishing.energyinst.org.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4406	-	Hydraulic fluid power — Fluids — Method for coding the level of contamination by solid particles
ISO 11171	-	Hydraulic Fluid Power – Calibration of Automatic Particle Counters for Liquids

(Copies of these documents are available from https://www.iso.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The fluids shall consist completely of hydrocarbon compounds, except as otherwise specified herein.

3.2 Chemical and physical requirements. The product shall conform to the requirements as specified in Table I. Requirements in Table I are absolute and not subject to correction for tolerance of test methods. The finished calibrating fluid shall be homogeneous, visually free from water, sediment, or suspended matter and shall be clear and bright at the ambient temperature or at 21° C (70° F), whichever is higher.

3.3 Additives

3.3.1 Corrosion inhibitor/lubricity improver (CI/LI). If so specified by the procuring activity, a CI/LI conforming to MIL-PRF-25017 shall be blended into the calibration fluid by the contractor. The amount added shall be equal to or greater than the minimum effective concentration listed in QPL-25017. The supplier may add any one of the CI/LI listed on the latest revision of QPL-25017. The supplier shall maintain documentation that the CI/LI used is an approved QPL-25017 product.

3.3.2 Antioxidants. If so specified by the procuring activity, an antioxidant additive shall be blended into the Type II calibrating fluid in total concentration of not less than 12.0 milligrams of active ingredient per liter (4.2 pounds of inhibitor, not including weight of solvents, per 1000 barrels of fluid) nor more than 24.0 milligrams of active ingredient per liter (8.4 pounds per 1000 barrels), in order to prevent the formation of gums and peroxides. The following antioxidant formulations or combinations thereof are approved for use:

a. 2,6-di-tert-butyl-4-methylphenol

b. 6-tert-butyl-2,4-dimethylphenol

c. 2,6-di-tert-butylphenol

d. 75 percent min-2,6-di-tert-butylphenol
25 percent max tert-butylphenols and tri-tert-butylphenols

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e. 72 percent min 6-tert-butyl-2,4-dimethylphenol
28 percent max tert-butyl-methylphenols and tert-butyl-dimethylphenols

f. 55 percent min 2,4-dimethyl-6-tert-butylphenol and
15 percent min 2,6-di-tert-butyl-4-methylphenol and
30 percent max mixed methyl and dimethyl tert-butylphenols

3.4 Workmanship. The finished calibrating fluid shall be homogeneous, visually free from undissolved water, sediment, or suspended matter and shall be clear and bright at the ambient temperature or at 21°C (70°F), whichever is higher.

3.5 Toxicity. The finished calibrating fluid shall have no adverse effect on the health of personnel when used for its intended purpose. The fluid shall contain no components that produce noxious vapors in such concentrations that would cause physical irritation to personnel during use or formulation under conditions of adequate ventilation. Percent composition of benzene shall be less than 0.01% of the total volume of the calibrating fluid due to the toxic properties of benzene.

TABLE I. Properties of finished fluid.

REQUIREMENTS	Type I	Type II	Type III	ASTM Method
Workmanship	Pass	Pass	Pass	Section 3.4
Color, Saybolt, lighter than	+25		+25	D156 [1], D6045
Total Acid Number, mg KOH/g		max 0.015		D3242
Aromatics, volume percent		max 20.0		D1319 [1], D6379
Olefins, volume percent		max 5.0		D1319
Benzene, volume percent	max 0.01	max 0.01	max 0.01	D3606 [2]
Sulfur, Mercaptan, mass percent or Doctor Test [3]		max 0.001 Sweet		D3227 [1] D4952
Flash Point, °C (°F)		min 38 (100)	min 79 (175)	D56[1], D93, D3828
Specific Gravity, kg/L at 15.6°C/15.6°C (60°F/60°F)	0.699 ± 0.002	0.770 ± 0.005	0.780 ± 0.005	D1298, D4052 [1], D7777
Freezing Point, °C (°F)			max -54 (-65)	D2386 [1], D5972, D7153, D7154
Viscosity, mm^2/s at				D445
0°C (32°F)	0.785 ± 0.01			
25°C (77°F)		1.17 ± 0.05		
37.8°C (100°F)	0.54 ± 0.01		2.47 ± 0.10	

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TABLE I. Properties of finished fluid - Continued.

REQUIREMENTS	Type I	Type II	Type III	ASTM Method
Distillation temperature, °C (°F)				D86
Initial boiling point, °C (°F)		min 149 (300)	min 216 (420)	
10 percent recovered °C (°F)			Report	
50 percent recovered, °C (°F)			221-232 (430-450)	
90 percent recovered, °C (°F)			Report	
Final Boiling Point, °C (°F)		max 210 (410)	232-246 (450-475)	
Recovery, volume percent		min 98.5		
Range, 5 to 95% points, °C (°F)	1.7 (3) [4]			
Residue, volume percent			max 1.5	
Loss, volume percent			max 1.5	
Acidity, Distillation Residue			Neutral	D1093
Vapor pressure at 37.8°C (100°F), kPa (psi)	max 13.8 (2.0)			D323, D 5191 [1]
Existent gum, mg/100 mL	max 2	max 5		D381 [5], IP 540 [1]
Potential gum, mg/100 mL	max 5.0			D873 [1, 6], D5304
Copper strip corrosion, 3 hr at 100°C, Rating	max No. 1	max No. 1	max No. 1	D130
Particulate Matter [7]				
Gravimetric, mg/L		max 2.0		D2276, D5452 [1]
or				
Particle counting, cumulative channel counts [8]				IP 565, IP 577, or D7619 [1]

Notes:

1.	Referee Test Method.
2.	Procedure B (Packed Gas Chromatographic Columns) in ASTM D3606 shall be the referee procedure.
3.	If the Doctor Test results in a failure ('positive' result), then Mercaptan sulfur content shall be determined by the referee test method ASTM D3227.
4.	Must Include Temperature of 98°C (208°F).
5.	Air Jet Method.
6.	5-Hour Aging Period.
7.	Particulate counting shall be the referee particulate matter procedure.

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TABLE I. Properties of finished fluid - Continued.

Notes:

8. Particle counting, where equipment and laboratory capability exist, is the alternate to the gravimetric method. The limits provided below are for particulate matter for procurement requirements only. Should particle counting result in failure, the gravimetric method shall be the referee.

Particle counting, individual channel counts & ISO Code	Channel Counts	ISO Code [8a]	Test Method IP 565, IP 577, or D7619
≥ 4 µm (c) [8b]	[8c]	20	
≥ 6 µm (c) [8b]	[8c]	18	
≥ 14 µm (c) [8b]	[8c]	15	
≥ 30 µm (c) [8b]	[8c]	13	

8a. Per Table 1 "Allocation of scale numbers" of ISO 4406.

8b. (c) indicates that the equipment has been calibrated in accordance with ISO 11171.

8c. Report.

3.7 Limiting Values. The following applies to all specified limits in this performance specification: For the purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right–hand digit used in expressing the specification limit according to the Rounding Method of ASTM E29.

3.8 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, environmentally preferable, or bio-based materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspections. The inspections shall be classified as quality conformance inspections.

4.2 Quality conformance inspection. Inspections of individual lots shall serve as a basis for acceptance and shall consist of all the examinations and tests specified in section 3. Use the chemical and physical requirements and applicable test methods as specified in Table I for conformance testing.

4.3 Sample. Each sample shall be of sufficient size to conduct all the quality conformance tests as specified herein. Unless otherwise specified, the quality conformance tests shall be performed on each required sample.

4.4 Sampling. Sampling shall be in accordance with ASTM D4057 or ASTM D4177.

4.5 Government requested sample. When requested, a 4-liter (1-gallon) sample shall be forwarded to the laboratory designated by the procuring activity for testing as specified herein.

4.6 Rejection. Failure of any calibrating fluid sample to conform to any of the specification requirements shall be cause for rejection of the lot represented.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fluids covered by this specification are intended for use in the calibration of military aircraft fuel system components. Exercise caution to avoid prolonged contact with the skin and observe Occupational Safety and Health Administration (OSHA) guidelines. Questions pertaining to the toxic effects should be referred to the appropriate departmental medical service.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Type of fluid required (i.e., Type I, II, or III).

c. Facility where Government requested test sample should be sent, if applicable.

d. Quantity required, and size and type of containers required.

e. Packaging requirements (see 5.1).

f. Addition of corrosion inhibitor to the calibrating fluid, if applicable.

g. Addition of anti-oxidant additive to the calibrating fluid, if applicable.

6.3 Definitions.

6.3.1 Bulk Lot of Material. An indefinite quantity of a homogeneous mixture of material contained in one isolated tank or kettle which is greater than 208 liters (55 gallons) in size, or a quantity manufactured by a single plant run through the same processing equipment during one continuous operation not exceeding a 24–hour period.

6.3.2 Packaged Lot of Material. As an indefinite number of 208-liter (55-gallon) drums or smaller unit containers of identical size and type, filled with a homogeneous mixture of material manufactured by a single plant run through the same processing equipment during one continuous operation not exceeding a 24-hour period.

6.4 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type

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II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, *Shelf-Life Management Program: Program Administration*; and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.5 Subject term (key word) listing.

High flash point
Corrosion inhibitor
Antioxidant
Stoddard solvent

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL:

Custodians:
Army – AV
Navy – AS
Air Force – 68
DLA – GS

Review activities:
Army – EA, MD
Air Force – 11

Preparing activity:
Air Force – 68
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